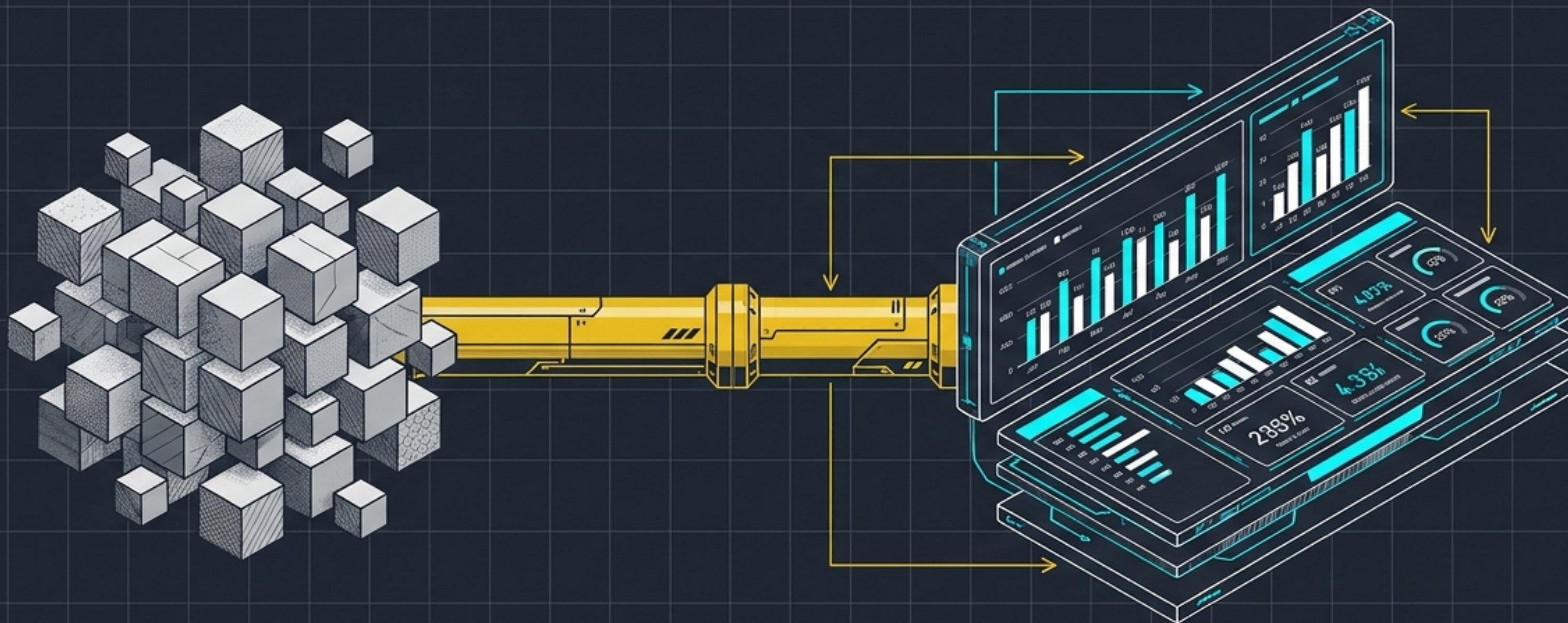


THE POWER BI BLUEPRINT

FROM STATIC SPREADSHEETS TO AUTOMATED DATA ENGINES



The Spreadsheet Era

Manual Copy/Paste
& **VLOOKUPS**



Maxes out,
slow to calculate



Static,
disconnected
charts

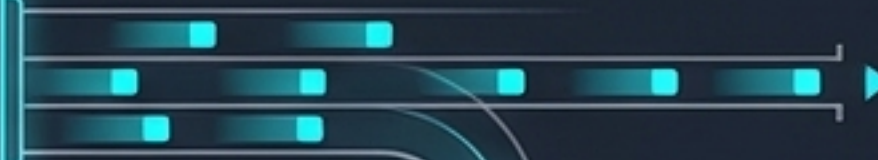


Emailed file
versions
(vFinal_v2)

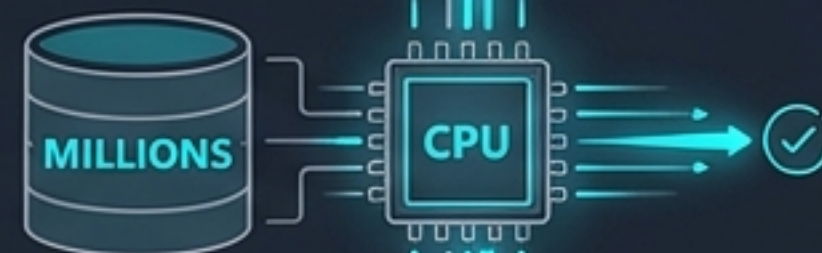


The BI Era

Automated
Data Pipelines



Handles **millions**
of rows **instantly**



Fully **interactive**,
cross-filtering
dashboards



Single source of
truth in the **Cloud**

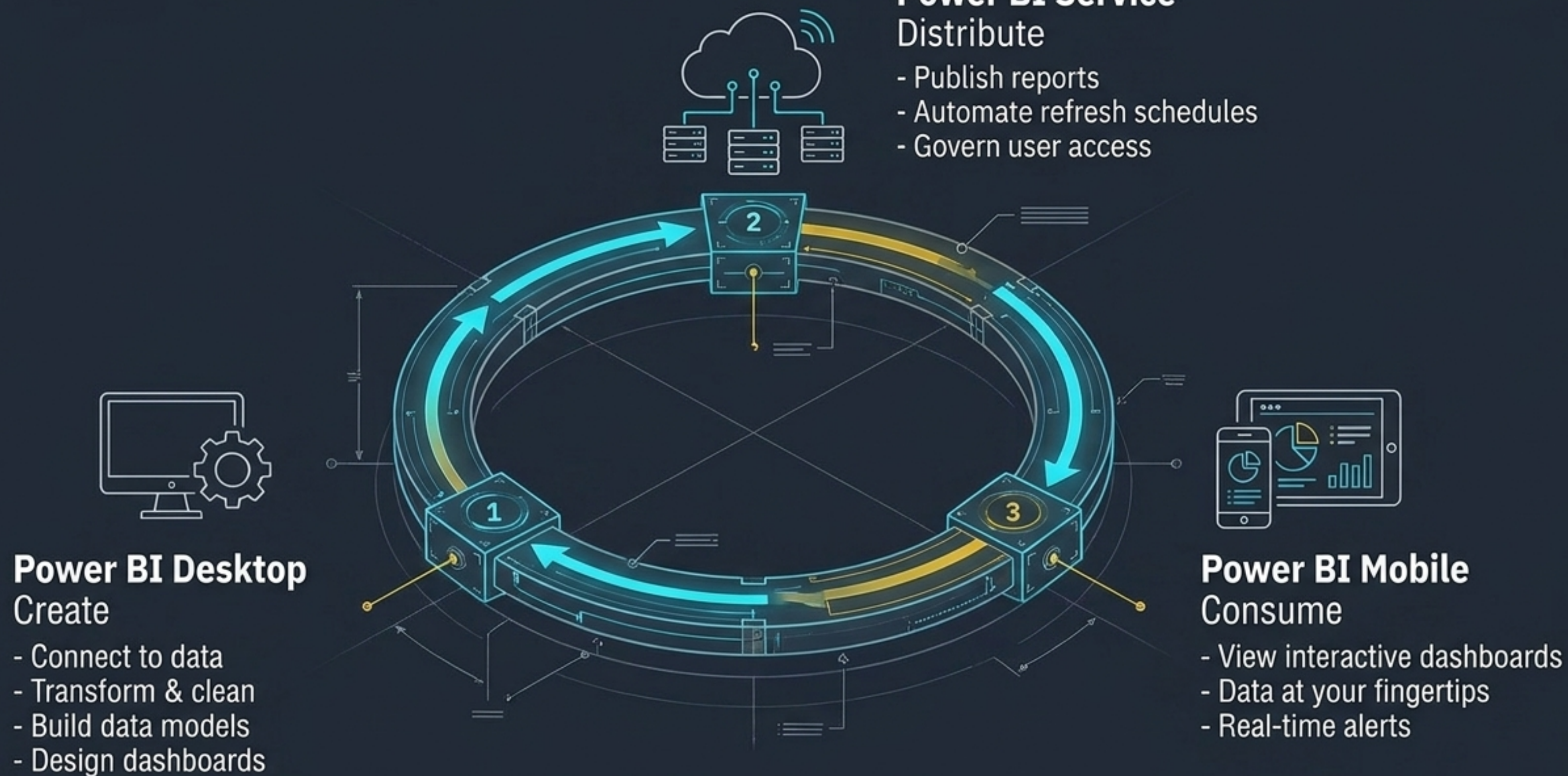


Stop putting out fires with repetitive weekly tasks.
Build a data engine once, and let it run.

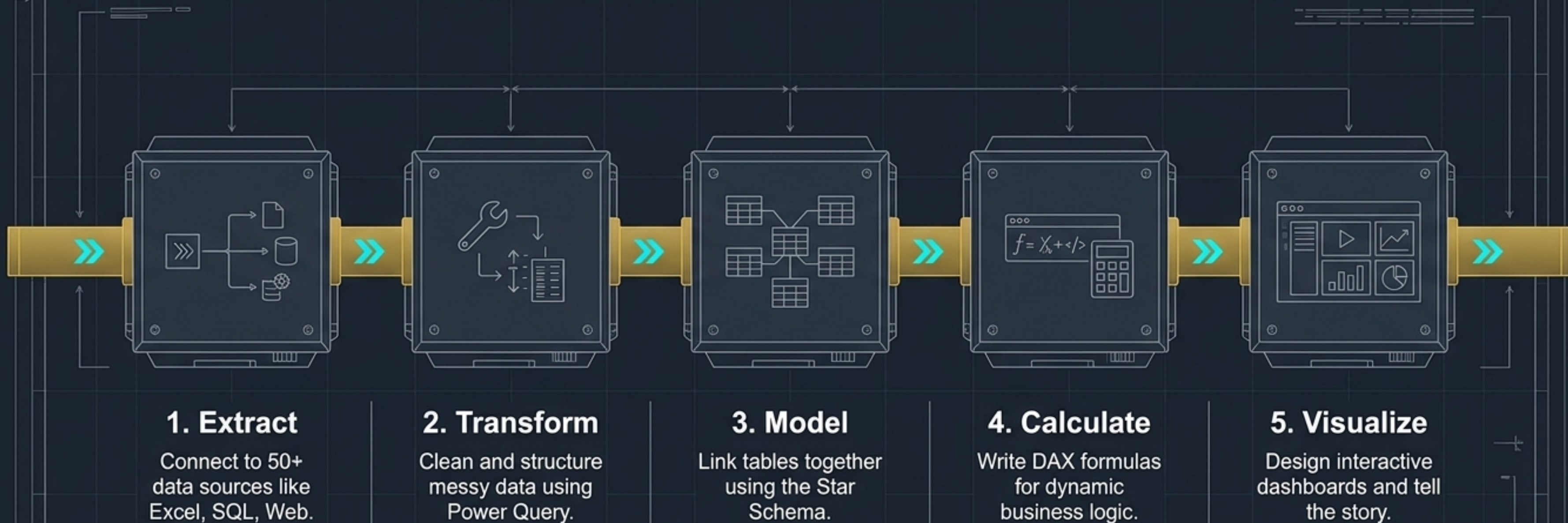


DIAGNOSTIC OUTPUT

The Ecosystem Triad



The Data Engine Pipeline



Phase 2: The Data Car Wash

Raw, Messy Data

Null values, unformatted text, system exports

The Power Query Engine

[Remove Blank Rows]

Structured Data

Ready for the data model

[Change Data Types]

[Promote Headers]

Power Query records your steps. When new data arrives tomorrow, you never repeat the work. Just click "Refresh".

Power Query in Action

Panel A: The Mess

	1	2
	Client_Info	Region_Data
1	Pinheiro, Vicente	France - Europe

[Split Column by Delimiter]



[Extract Text Between Delimiters]

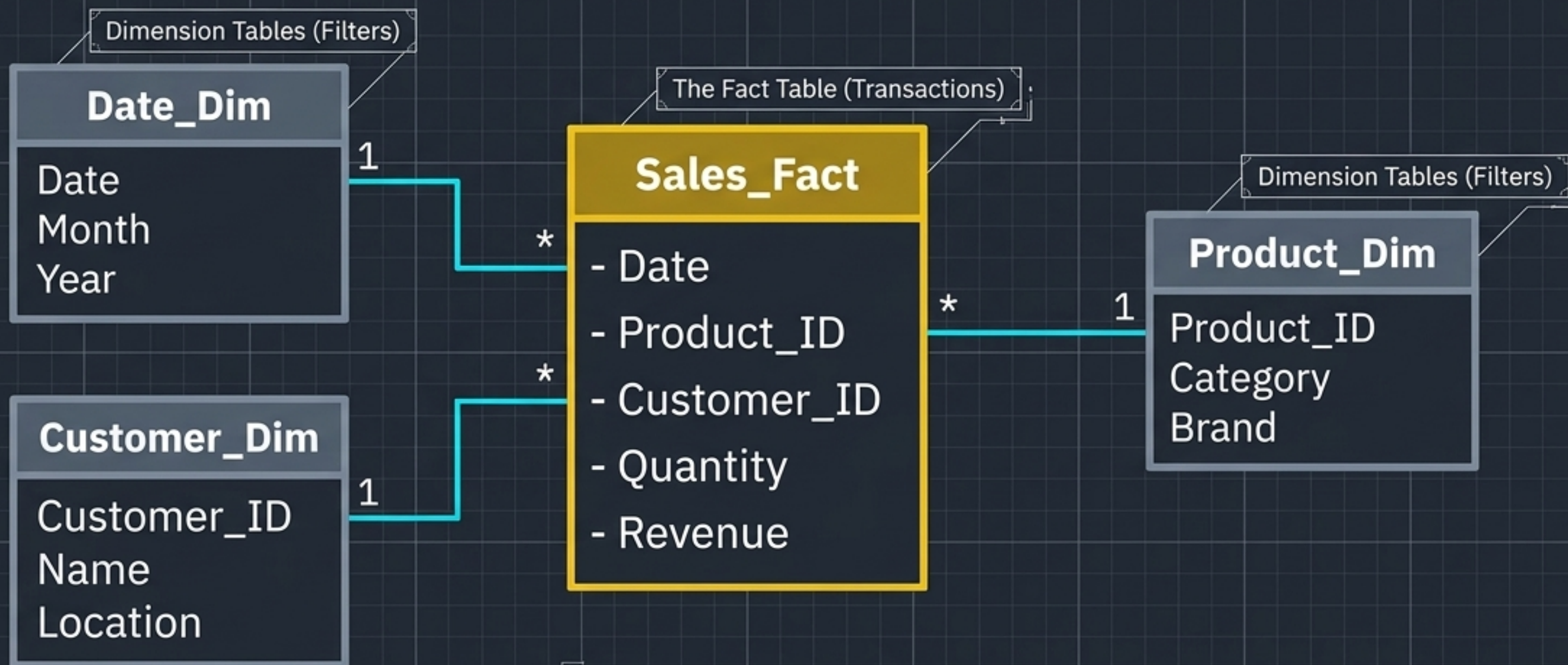
Panel B: The Fix

	1	2	3	4
	First Name	Last Name	Country	Continent
1	Vicente	Pinheiro	France	Europe



The Magic Wand: Use the 'Column From Examples' feature to let AI detect your text patterns and auto-fill the transformations.

Phase 3: The Star Schema



Stop using VLOOKUP to build massive, slow tables. Use 1-to-Many relationships to link unique IDs (e.g., Product_ID) to your transaction history instantly.

Phase 4: Demystifying DAX

```
Margin % = DIVIDE ( [Total Margin], SUM(Sales[Revenue]) )
```

The Measure Name:
Your custom metric.

The Safe Operator:
Handles division by zero errors automatically, unlike standard Excel.

The Numerator:
Reusing a previously built measure to keep code clean and modular.

The Aggregation:
Calculates dynamically based on what the user clicks on the dashboard.

DAX answers one question:

Given the current filters on the page, what number should I return?

The Calculation Matrix



Calculated Columns

How it works

Calculates row-by-row during data refresh.

Resource Cost

Increases file size. Takes up physical RAM storage.

When to use

To slice, filter, or categorize data (e.g., creating a 'Profit Band' column to use in a chart axis).



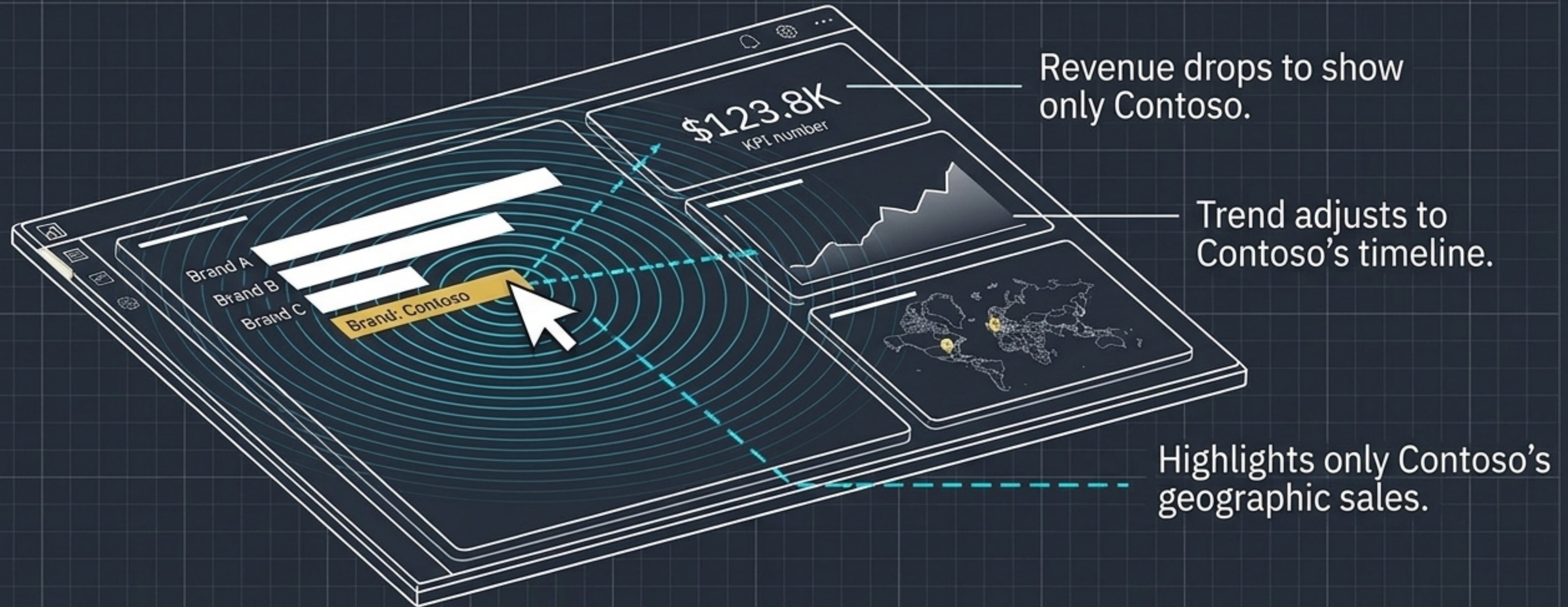
Measures (DAX)

Calculates on-the-fly based on user filter context.

Takes zero storage space.
Uses CPU only when viewed.

For aggregate numbers (e.g., Total Revenue, YTD Sales, Profit Margin %).

Phase 5: The Filter Context Ripple



The Filter Context: In Power BI, every visual is a filter. Clicking one data point creates a ripple effect, recalculating every measure on the page instantly.

The Z-Pattern of Storytelling

1. High-Level KPIs

The most critical numbers (Revenue, Margin). The first thing the eye naturally sees.

3. Categorical Breakdowns

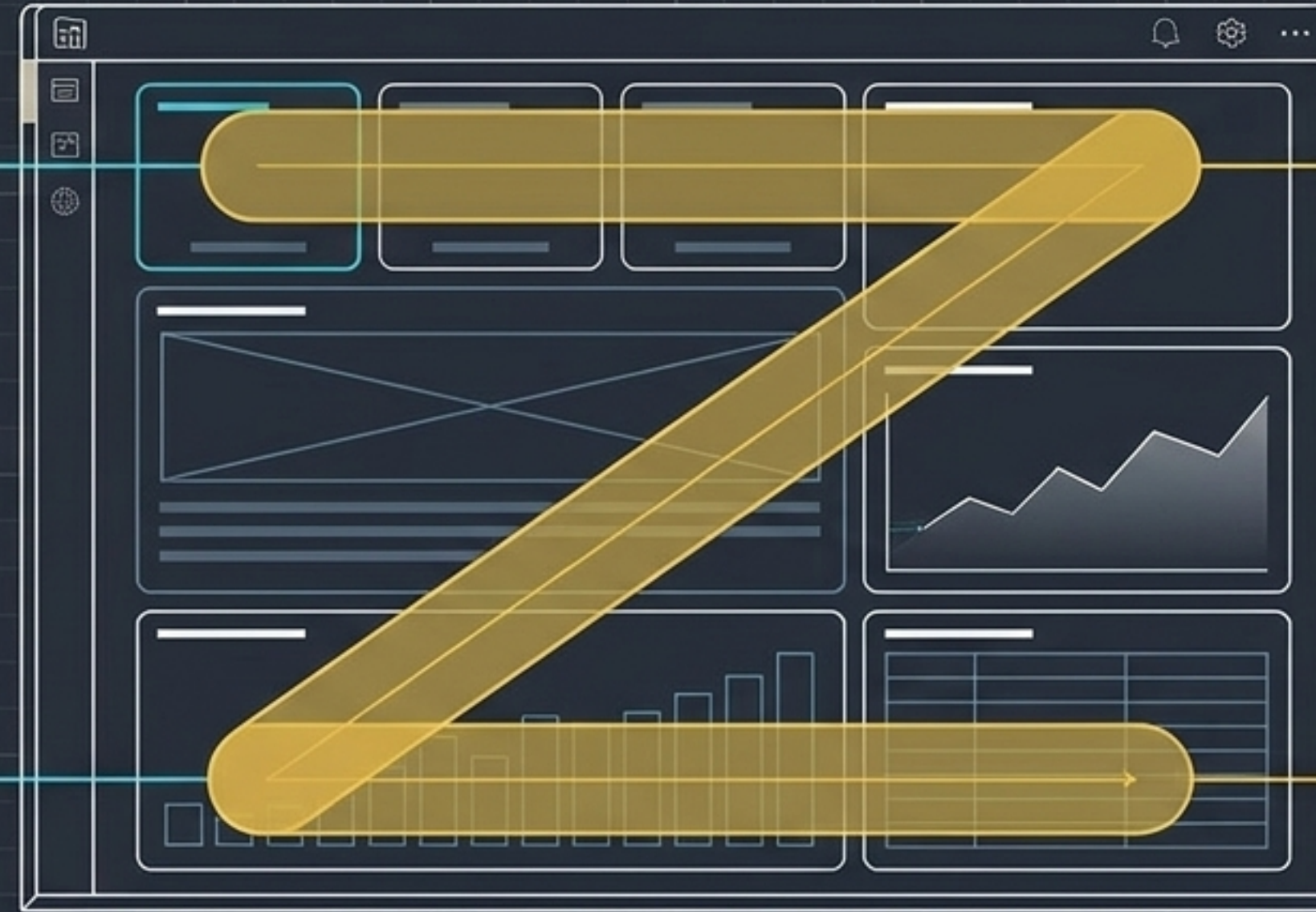
Distributions and rankings (Bar charts showing sales by product or region).

2. Time/Trend Analysis

Contextualizing the KPIs over time (Line charts and historical context).

4. Granular Details

The final destination for the eye. Deep-dive data tables and specific matrices.



Don't place visuals randomly. Guide the user's eye naturally from macro-summary to micro-detail.

The Studio Polish

Amateur Execution



Professional Execution



3 Pro-Tips

1. The Background Hack

Build your structural layout (shapes, shadows, headers) in Figma or PowerPoint, export as a PNG, and upload it as the Power BI Canvas Background.

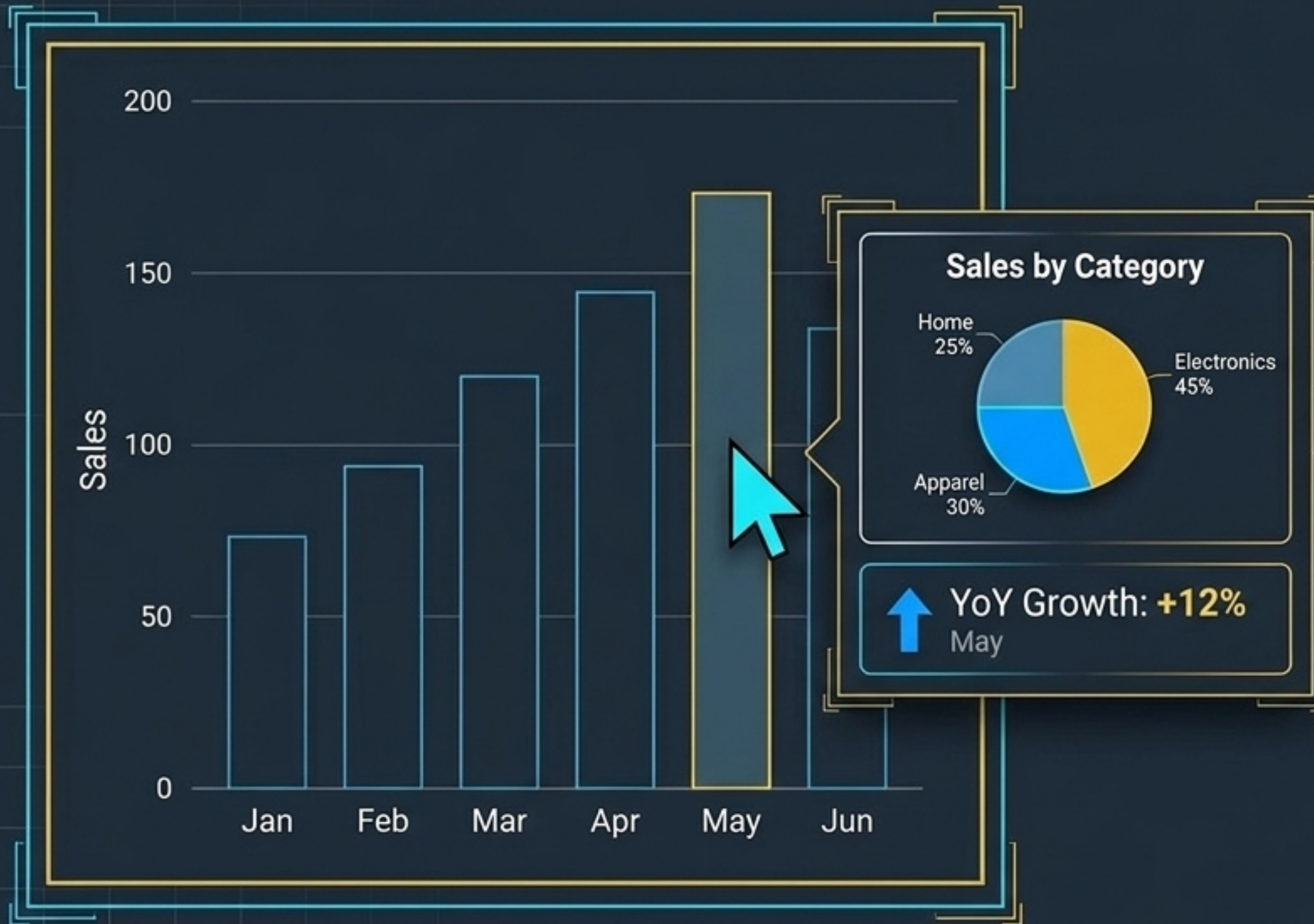
2. Total Transparency

Turn off the default white background on all Power BI visuals so they blend seamlessly into your custom canvas.

3. Color Harmony

Use exact hex codes (e.g., #1A2B3C) for brand matching. Avoid default palettes; rely on monochromatic gradients for continuous data.

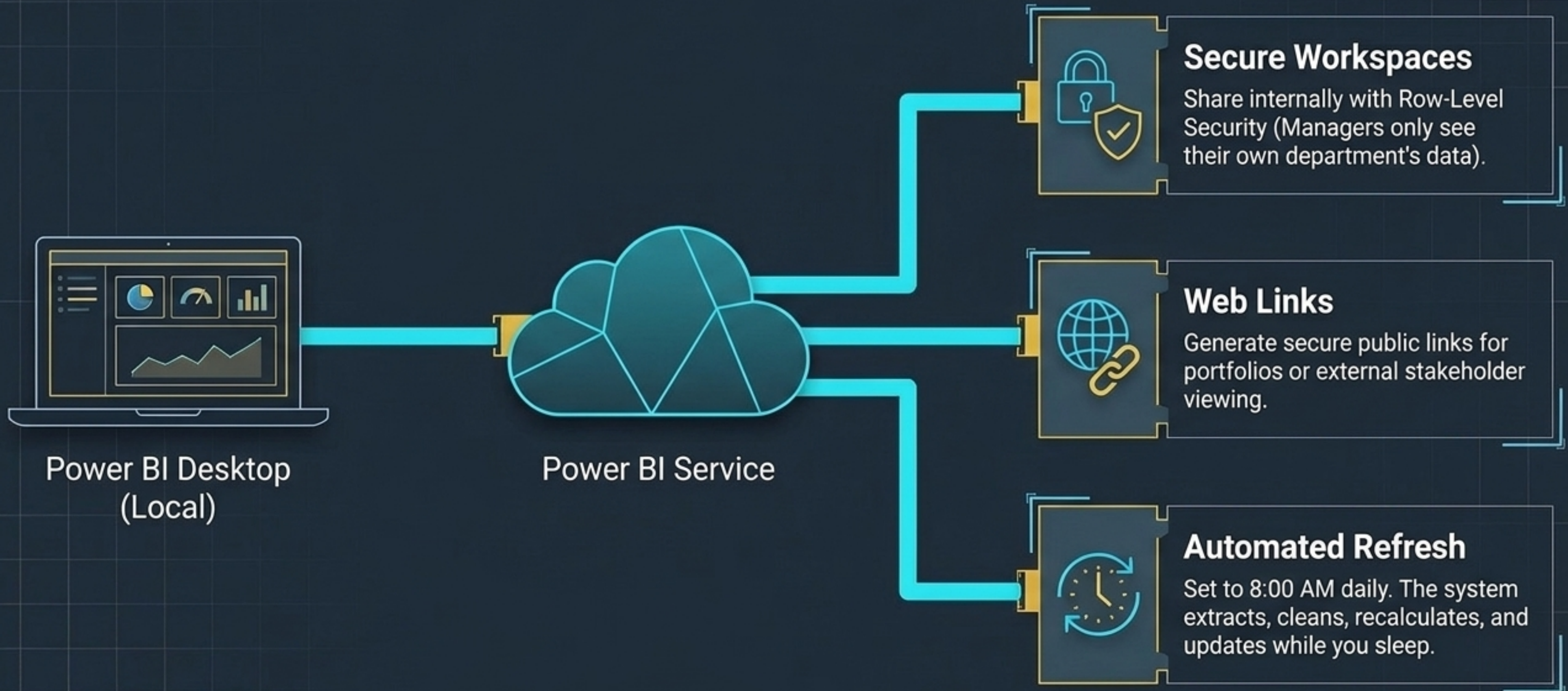
Layering Information: Tooltips



The Hover Effect

Embed custom mini-dashboards inside your main dashboard. Answer the next logical question before the user even has to ask it, without crowding your primary page.

Phase 6: Distribute & Automate



The Anatomy of a World-Class Dashboard

